

SIEGE ENGINES AND EARLY CANNON

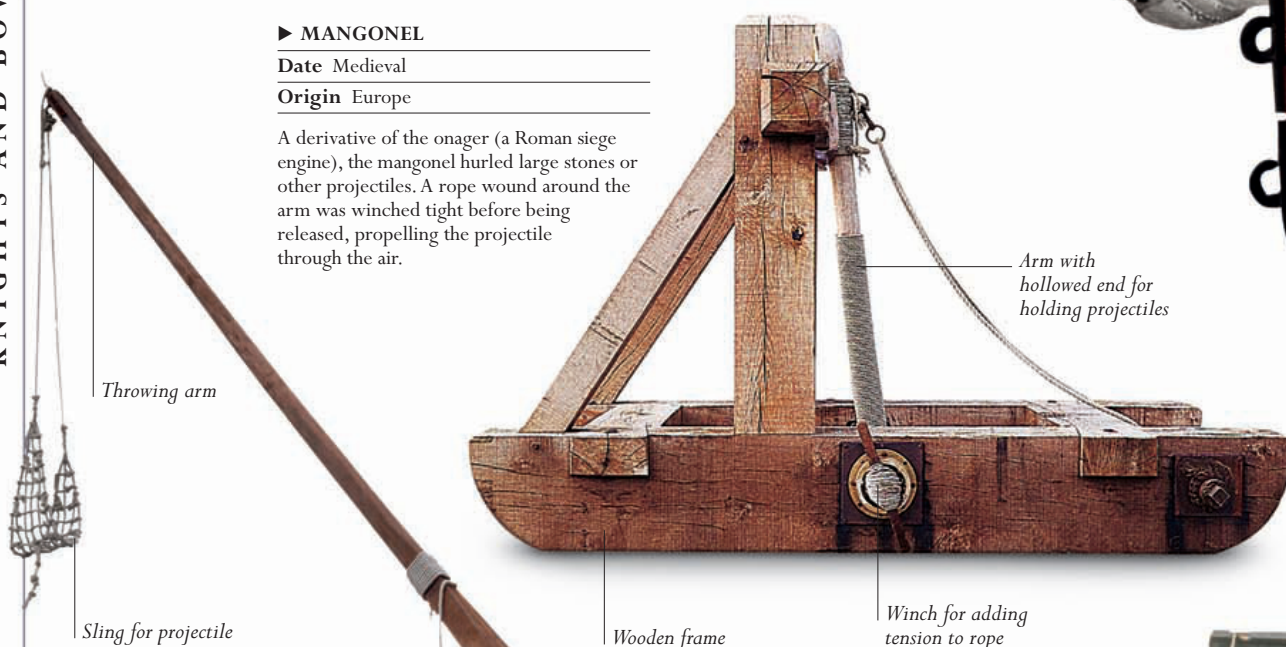
During the Middle Ages, sieges were a far more common way of conducting warfare than open battle. Powerful siege engines that could batter fortification walls were used, allowing besieging armies to breach enemy defences. The development of gunpowder weapons from the early 14th century onwards increased the psychological aspect of siege warfare – the threat of devastation from this new form of firepower could drastically shorten a siege.

► MANGONEL

Date Medieval

Origin Europe

A derivative of the onager (a Roman siege engine), the mangonel hurled large stones or other projectiles. A rope wound around the arm was winched tight before being released, propelling the projectile through the air.

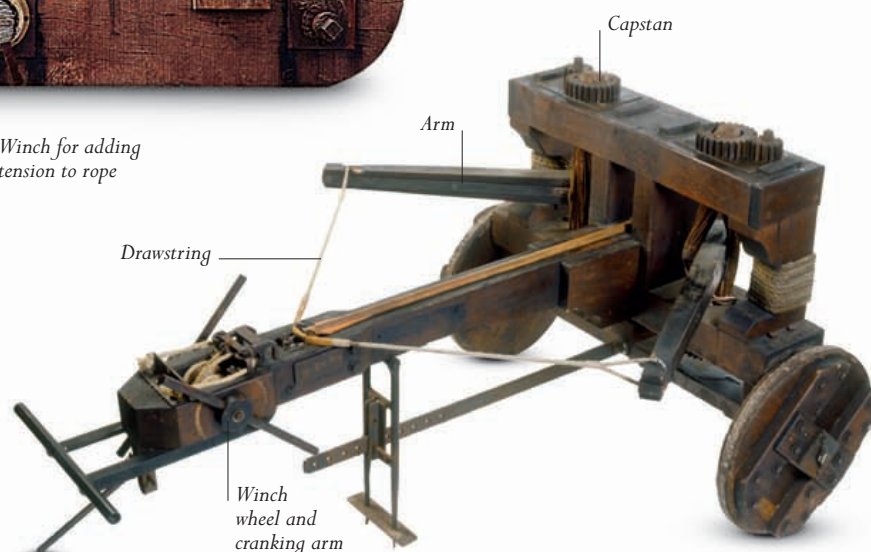
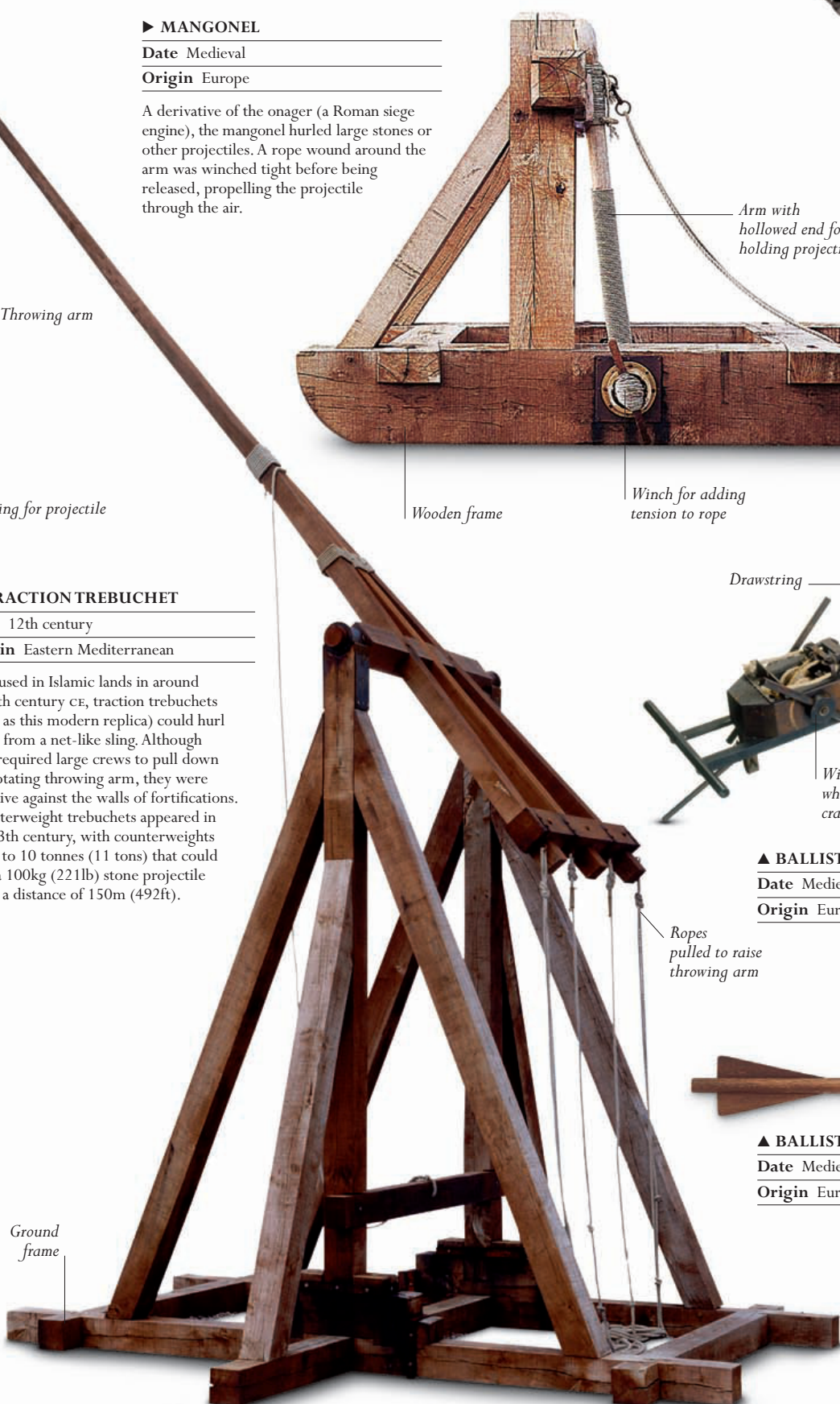


► TRACTION TREBUCHET

Date 12th century

Origin Eastern Mediterranean

First used in Islamic lands in around the 7th century CE, traction trebuchets (such as this modern replica) could hurl rocks from a net-like sling. Although they required large crews to pull down the rotating throwing arm, they were effective against the walls of fortifications. Counterweight trebuchets appeared in the 13th century, with counterweights of up to 10 tonnes (11 tons) that could hurl a 100kg (221lb) stone projectile up to a distance of 150m (492ft).



▲ BALLISTA

Date Medieval

Origin Europe

Based on a weapon first developed in Greek and Roman times, the ballista was a giant crossbow with a bolt that was operated by cranking back a drawstring and then releasing the tension with a lever. It was primarily an antipersonnel weapon, rather than one designed to damage fortifications.



▲ BALLISTA BOLT

Date Medieval

Origin Europe

The bolts fired by a ballista catapult travelled with such force that a single bolt could pierce three men at once. The bolts were not so effective in damaging masonry however.

